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# Fast verification of fault tolerant magic state preparation circuits

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# Magic states and resource estimation

- Often the most expensive resource to prepare
- Dominates most resource estimates
- Optimising the bottleneck is the best way to improve performance

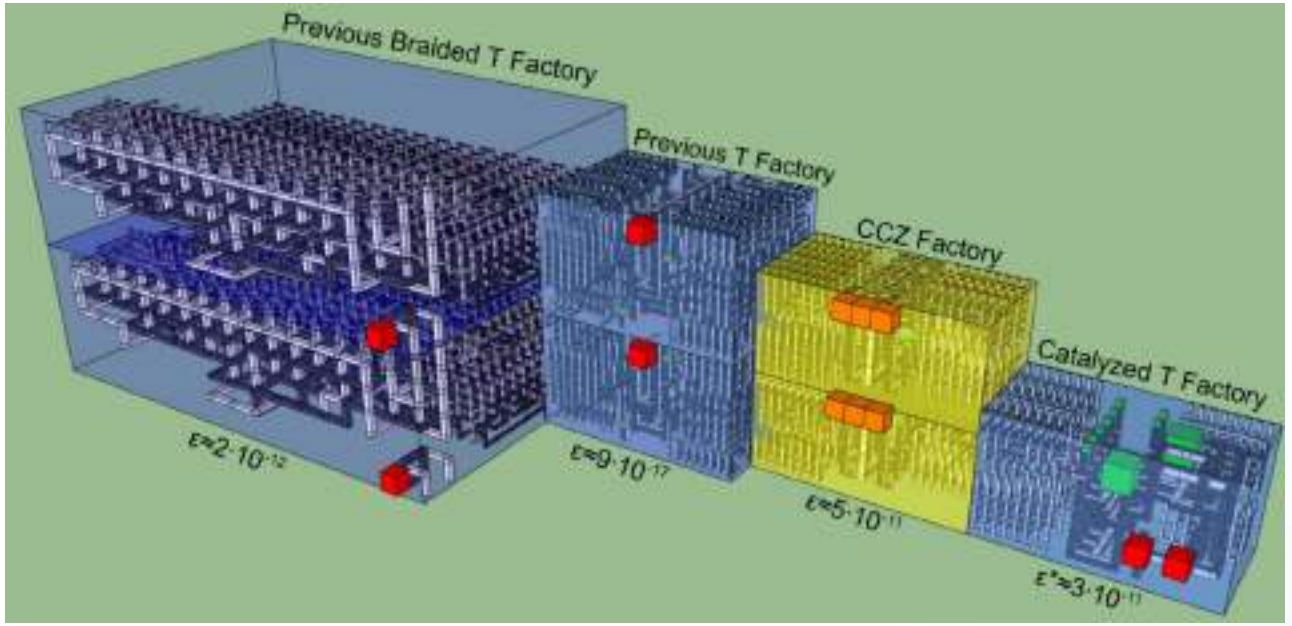
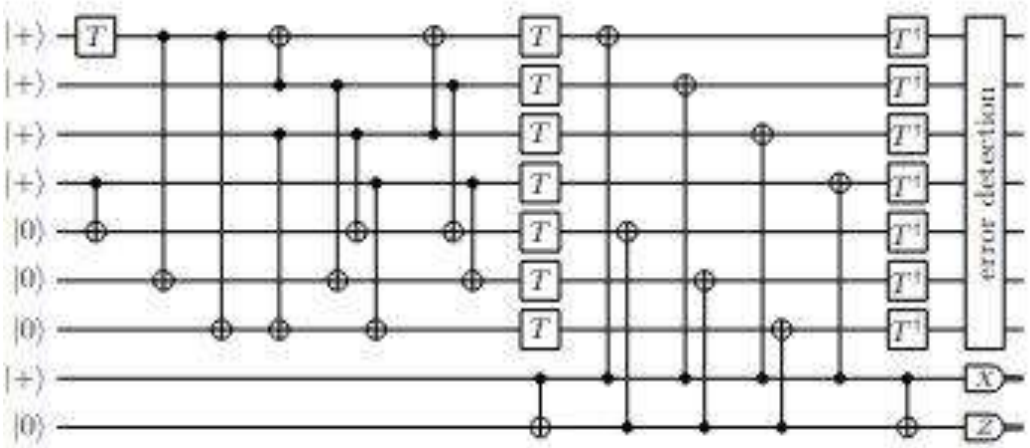



Figure 2, doi:10.22331/q-2019-04-30-135, Gidney and Fowler, 2019


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# Why magic state preparation is hard

- Number of fault combinations
- Non-Clifford circuits
- Checking the goodness of an error





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# Surmounting the difficulties: limit the gate set

- CNOT + T circuits are powerful enough to prepare magic states
- Yet simple enough to reason about fault propagation
- Pauli faults propagate to Clifford (DCP) output errors
- Storing and multiplying output errors is efficient


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# Checking the goodness of an output error

- Use lookup tables to calculate the weight of an error – practical for many use cases

- Can efficiently expand a DCP error as a linear combination of Paulis and check each one


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# Comb

- Implemented all of this in a C library
- Small interface: one FFI call
- Input: a list of output errors and fault locations, description of the magic state being prepared
- Output: whether the circuit is fault tolerant or not
- Throughput: ~ one million fault combinations per core per minute for the [[17, 1, 5]] colour code
- Verification of higher distance magic state preparation circuits well within reach


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# Demonstration

- Can verify the fault tolerance of a distance-five magic state preparation circuit within minutes on a single CPU core (!)

- This circuit was discovered with Comb


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# Conclusion and future directions

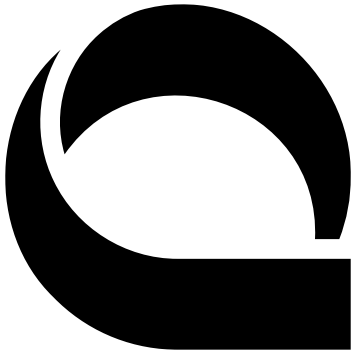
- Paper and code on the way
- Many more circuits to be discovered
- Scale to distance-seven with HPC
- Partial fault tolerance



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# Thank you

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